

discuss both theoretical and Practical Implementation from Scratch.

This chapter spans 5 parts:

1. What is an Artificial Neural Network?
2. Types of Artificial Neural Network.
3. Activation Functions and Their types?
4. How Does Backpropagation work?
5. Practical Implementation of ANN in Keras and Tensorflow.

1. What is an Artificial Neural Network?

Artificial neural networks are one of the main tools used in machine learning. As the “neural” part of their name suggests, they are brain-inspired systems that are intended to replicate the way that we humans learn. Neural networks consist of input and output layers, as well as (in most cases) a hidden layer consisting of units that transform the input into something that the output layer can use. They are excellent tools for finding patterns that are far too complex or numerous for a human programmer to extract and teach the machine to recognize.

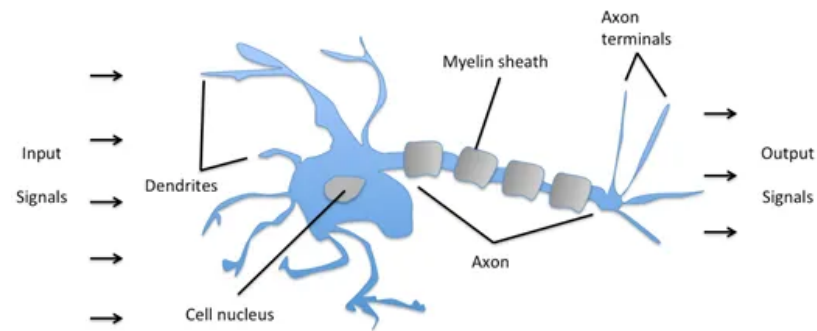
While neural networks (also called “perceptrons”) have been around since the 1940s, it is only in the last several decades where they have become a major part of artificial intelligence. This is due to the arrival of a technique called “backpropagation,” which allows networks to adjust their hidden layers of neurons in situations where the outcome doesn’t match what the creator is hoping for — like a network designed to recognize dogs, which misidentifies a cat, for example.

Another important advance has been the arrival of deep learning neural networks, in which different layers of a multilayer network extract different features until it can recognize what it is looking for.

1.1: Basic Structure of ANNs

The idea of ANNs is based on the belief that working of the human brain by making the right connections can be imitated using silicon and wires as living **neurons** and **dendrites**.

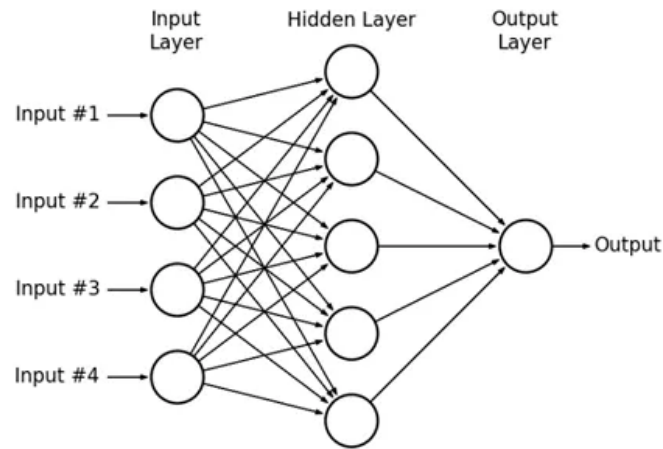
The human brain is composed of 86 billion nerve cells called **neurons**. They are connected to other thousand cells by **Axons**. Stimuli from the external environment or inputs from sensory organs are accepted by dendrites. These inputs create electric impulses, which quickly travel through the neural network. A neuron can then send the message to another neuron to handle the issue or does not send it forward.



Schematic of a biological neuron.

ANNs are composed of multiple **nodes**, which imitate biological **neurons** of the human brain. The neurons are connected by links and they interact with each other. The nodes can take input data and perform simple operations on the data. The result of these operations is passed to other neurons. The output at each node is called its **activation** or **node value**.

Each link is associated with **weight**. ANNs are capable of learning, which takes place by altering weight values. The following illustration shows a simple ANN –

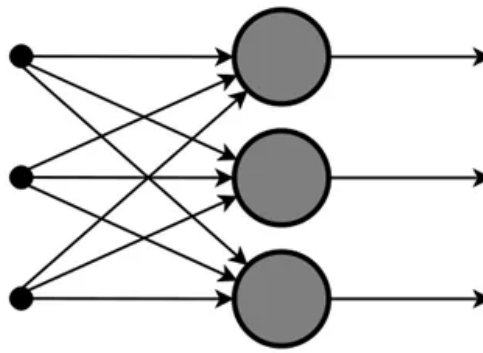


2. Types of Artificial Neural Networks

There are two Artificial Neural Network topologies – **FeedForward** and **Feedback**.

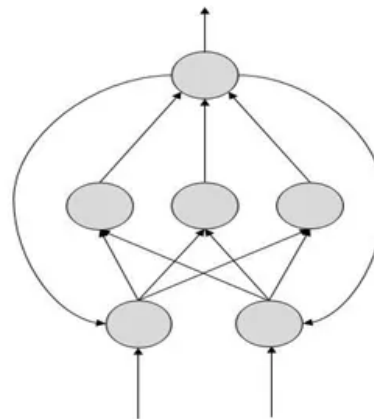
2.1: FeedForward ANN

In this ANN, the information flow is unidirectional. A unit sends information to another unit from which it does not receive any information. There are no feedback loops. They are used in pattern generation/recognition/classification. They have fixed inputs and outputs.



2.2: FeedBack ANN

Here, feedback loops are allowed. They are used in content-addressable memories.



3. Activation Functions and There Types?

3.1: What is Activation Function?

It's just a thing function that you use to get the output of the node. It is also known as **Transfer Function**.

Activation functions are really important for an Artificial Neural Network to learn and make sense of something reallocated and Non-linear complex

functional mappings between the inputs and response variable. *They introduce non-linear properties to our Network. Their main purpose is to convert an input signal of a node in an ANN to an output signal.* That output signal now is used as an input in the next layer in the stack.

Specifically in A-NN we do the sum of products of inputs(X) and their corresponding Weights (W) and apply an Activation function $f(x)$ to it to get the output of that layer and feed it as an input to the next layer.

3.2: Types of activation Functions?

It is used to determine the output of neural network like yes or no. It maps the resulting values in between 0 to 1 or -1 to 1 etc. (depending upon the function).

The Activation Functions can be based on 2 types-

1. Linear Activation Function
2. Non-linear Activation Functions

Linear Activation Function

As you can see the function is a line or linear. Therefore, the output of the functions will not be confined between any range.

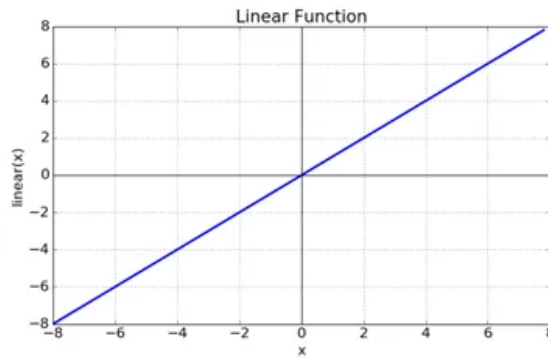


Fig: Linear Activation Function

Equation: $f(x) = x$

Range: (-infinity to infinity)

It doesn't help with the complexity of various parameters of usual data that is fed to the neural networks.

Non-linear Activation Function

The Nonlinear Activation Functions are the most used activation functions.

Nonlinearity helps to makes the graph look something like this

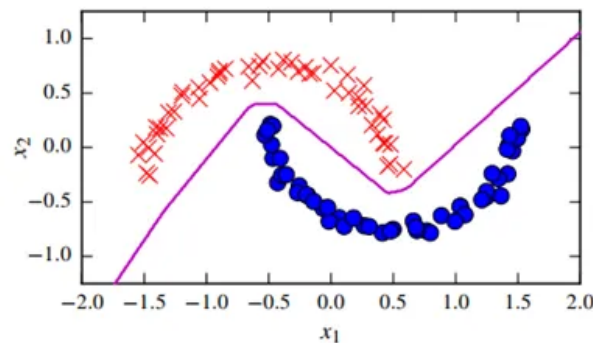
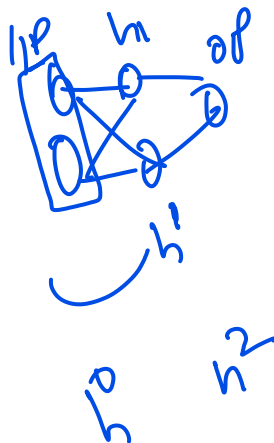


Fig: Non-linear Activation Function

$$l = [1, 2, 3, 4, \dots]$$

$$[w^1]_{1 \times \text{hidden}}$$

$$b^1 = [\cdot]_{1 \times 3}$$

$$h^0 = x$$

$$z^l = h^{l-1} w^l + b^l$$

It makes it easy for the model to generalize or adapt to a variety of data and to differentiate between the outputs.

The main terminologies needed to understand for nonlinear functions are:

Derivative or Differential: Change in y-axis w.r.t. change in the x-axis. It is also known as a slope.

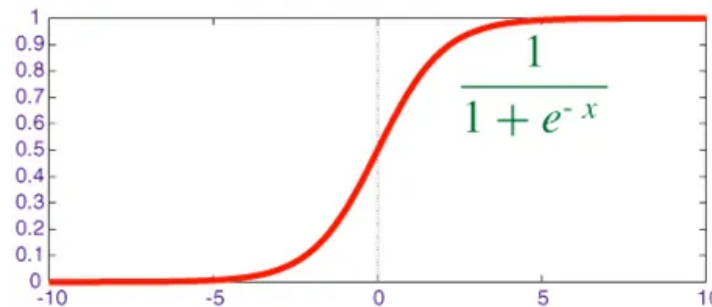
Monotonic function: A function that is either entirely non-increasing or non-decreasing.

The Nonlinear Activation Functions are mainly divided on based on **range** or **curves**-

Different Types of Activation function in non-Linear

1. Sigmoid Activation Function

The Sigmoid Function curve looks like an S-shape.



Sigmoid Function

The main reason why we use the sigmoid function is that it exists between (0 to 1). Therefore, it is especially used for models where we have to **predict the**

$$\begin{aligned} z' &= h^0 w' + b' \\ z' &= x w' + b \\ h &= \sigma^2(z') \\ h' &= \sigma'(z') \\ h' &= \sigma'(x w' + b) \end{aligned}$$

probability as an output. Since the probability of anything exists only between the range of **0 and 1**, sigmoid is the right choice.

The function is **differentiable**. That means, we can find the slope of the sigmoid curve at any two points.

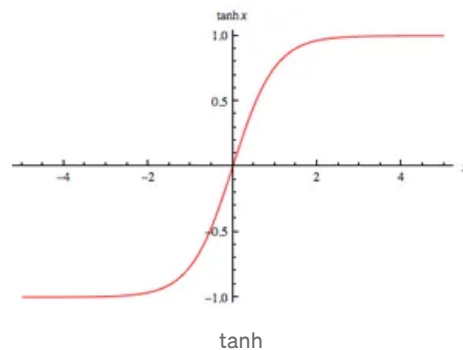
The function is **monotonic** but the function's derivative is not.

The logistic sigmoid function can cause a neural network to get stuck at the training time.

The **softmax function** is a more generalized logistic activation function that is used for multiclass classification.

2. Tanh Activation Function

tanh is also like logistic sigmoid but better. The range of the tanh function is from (-1 to 1). tanh is also sigmoidal (s-shaped).



The advantage is that the negative inputs will be mapped strongly negative and the zero inputs will be mapped near zero in the tanh graph. The function is **differentiable**.

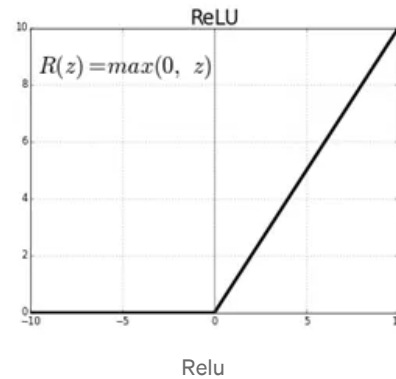
The function is **monotonic** while its **derivative is not monotonic**.

The tanh function is mainly used classification between two classes.

Both tanh and logistic sigmoid activation functions are used in feed-forward nets.

3. ReLU (Rectified Linear Unit) Activation Function

The ReLU is the most used activation function in the world right now. Since it is used in almost all the convolutional neural networks or deep learning.



As you can see, the ReLU is half rectified (from bottom). $f(z)$ is zero when z is less than zero and $f(z)$ is equal to z when z is above or equal to zero.

Range: [0 to infinity)

The function and its derivative **both are monotonic**.

But the issue is that all the negative values become zero immediately which decreases the ability of the model to fit or train from the data properly. That means any negative input given to the ReLU activation function turns the

value into zero immediately in the graph, which in turn affects the resulting graph by not mapping the negative values appropriately.

4: How Backpropagation works?

Let's dive into the mathematics part and see the backpropagation working with feedforward and feed backward Neural Network in depth.

Approach

Step 1: Randomly initialize the Weights to a small number close to 0 (but not 0).

Step 2: Input the first observation of your dataset in the input layer, each feature in one input node.

Step 3: Forward-Propagation: from left to right, the neurons are activated in a way that the impact of each neuron's activation is limited by the weights. Propagates the activations until getting the predicted result y .

Step 4: Compare the predicted result to the actual result. Measure the generated error.

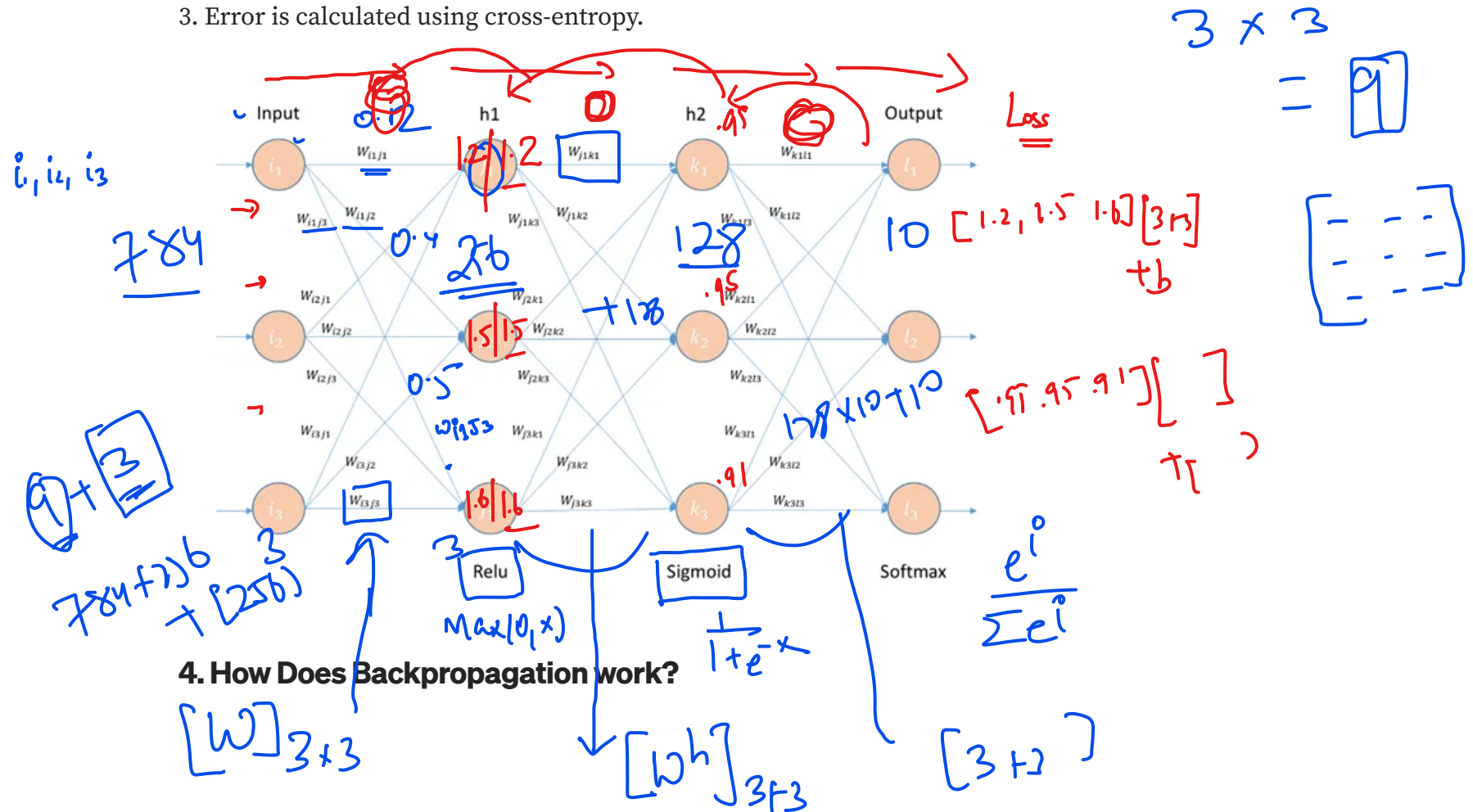
Step 5: Back-Propagation: from left to right, the error is back-Propagated. Update the weights according to how much they are responsible for the error. The Learning rate decides how much we update the weights.

Step 6: Repeat step 1 to step 5 and updates the weights after each observation.

Step 7: When the whole training set passed through the ANN that makes an epoch. Redo more epoch.

Architecture

1. Build a Feed-Forward neural network with 2 hidden layers. All the layers will have 3 Neurons each.
2. 1st and 2nd hidden layers will have RELU and sigmoid respectively as activation functions. The final layer will have a Softmax activation function.
3. Error is calculated using cross-entropy.



4. How Does Backpropagation work?

Inputs

$$\text{Inputs} = [i_1 \quad i_2 \quad i_3]$$

$$= [0.1 \quad 0.3 \quad 0.5]$$

$$\begin{bmatrix} 0.1 \\ 0.3 \\ 0.5 \end{bmatrix}$$

Actual Outputs

$$\text{Outputs} = [y_1 \quad y_2 \quad y_3]$$

$$= [0.0 \quad 1.0 \quad 0.0]$$

Randomly initialize the Weights to small number close to 0 (but not 0)

Weights for Layer 1

$$W_{ij} = \begin{bmatrix} W_{i1j1} & W_{i1j2} & W_{i1j3} \\ W_{i2j1} & W_{i2j2} & W_{i2j3} \\ W_{i3j1} & W_{i3j2} & W_{i3j3} \end{bmatrix}$$

$$W_{ij} = \begin{bmatrix} 0.2 & 0.4 & 0.5 \\ 0.3 & 0.7 & 0.1 \\ 0.2 & 0.6 & 0.9 \end{bmatrix} \quad 3 \times 3$$

input to first hidden layer

$$w^{\text{new}} = w^{\text{old}} - \eta \frac{\partial \text{loss}}{\partial w}$$

$$w^{\text{new}} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} - \begin{bmatrix} 1 & 3 & \end{bmatrix} \begin{bmatrix} \\ \\ \end{bmatrix}$$

Weights for Layer 2

$$W_{jk} = \begin{bmatrix} W_{j1k1} & W_{j1k2} & W_{j1k3} \\ W_{j2k1} & W_{j2k2} & W_{j2k3} \\ W_{j3k1} & W_{j3k2} & W_{j3k3} \end{bmatrix}$$

$$W_{jk} = \begin{bmatrix} 0.4 & 0.6 & 0.1 \\ 0.2 & 0.7 & 0.3 \\ 0.8 & 0.1 & 0.5 \end{bmatrix}$$

hidden 1 to hidden 2

Weights for Layer 3

$$W_{kl} = \begin{bmatrix} W_{k1l1} & W_{k1l2} & W_{k1l3} \\ W_{k2l1} & W_{k2l2} & W_{k2l3} \\ W_{k3l1} & W_{k3l2} & W_{k3l3} \end{bmatrix}$$

$$W_{kl} = \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.4 & 0.6 & 0.8 \\ 0.1 & 0.9 & 0.7 \end{bmatrix}$$

hidden 2 to output

✓

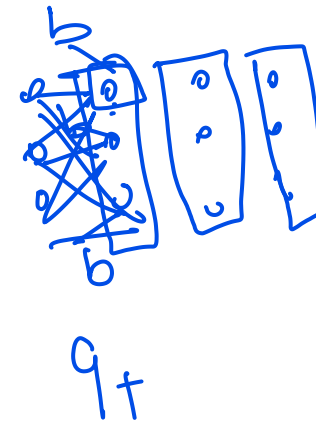
Feedforward Neural Network: Forward-Propagation: from left to right, the neurons are activated in a way that the impact of each neuron's activation is limited by the weights. Propagates the activations until getting the predicted result y.

Layer 1 Matrix Operation

input to hidden 1

$$= \underbrace{[i_1 \ i_2 \ i_3]}_{1 \times 3} \times \underbrace{\begin{bmatrix} W_{i1j1} & W_{i1j2} & W_{i1j3} \\ W_{i2j1} & W_{i2j2} & W_{i2j3} \\ W_{i3j1} & W_{i3j2} & W_{i3j3} \end{bmatrix}}_{3 \times 3} + \underbrace{[b_{j1} \ b_{j2} \ b_{j3}]}_{1 \times 3}$$

$1 \times 3 \quad 3 \times 3 \quad 1 \times 3 \rightarrow 1 \times 3$



$$Z_p = \sum W^T x$$

$$= \underbrace{[0.1 \ 0.3 \ 0.5]}_{1 \times 3} \times \underbrace{\begin{bmatrix} 0.2 & 0.4 & 0.5 \\ 0.3 & 0.7 & 0.1 \\ 0.2 & 0.6 & 0.9 \end{bmatrix}}_{3 \times 3} + \underbrace{[1.0 \ 1.0 \ 1.0]}_{1 \times 3}$$

$[0.2 \ 0.6 \ 0.5] + [1 \ 1 \ 1]$

Pre Activation

$$\begin{bmatrix} h_{1in1} & h_{1in2} & h_{1in3} \end{bmatrix} = \boxed{[1.2 \ 1.6 \ 1.5]}$$

RELU operation on layer 1

$$\text{RELU} = \max(0, x)$$

$$\begin{bmatrix} h1_{out1} & h1_{out2} & h1_{out3} \end{bmatrix} = \begin{bmatrix} \max(0, h1_{in1}) & \max(0, h1_{in2}) & \max(0, h1_{in3}) \end{bmatrix}$$

After Apply Actut

$$\begin{bmatrix} h1_{out1} & h1_{out2} & h1_{out3} \end{bmatrix} = \begin{bmatrix} 1.2 & 1.6 & 1.5 \end{bmatrix}$$

Layer 2 Matrix Operation

$$= \begin{bmatrix} h1_{out1} & h1_{out2} & h1_{out3} \end{bmatrix} \times \begin{bmatrix} W_{j1k1} & W_{j1k2} & W_{j1k3} \\ W_{j2k1} & W_{j2k2} & W_{j2k3} \\ W_{j3k1} & W_{j3k2} & W_{j3k3} \end{bmatrix} + \begin{bmatrix} b_{k1} & b_{k2} & b_{k3} \end{bmatrix}$$

$$= \begin{bmatrix} 1.2 & 1.6 & 1.5 \end{bmatrix} \times \begin{bmatrix} 0.4 & 0.6 & 0.1 \\ 0.2 & 0.7 & 0.3 \\ 0.8 & 0.1 & 0.5 \end{bmatrix} + \begin{bmatrix} 1.0 & 1.0 & 1.0 \end{bmatrix}$$

$$= \begin{bmatrix} 2.2 & 1.4 & 1.1 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

Pre activation

$$\begin{bmatrix} h2_{in1} & h2_{in2} & h2_{in3} \end{bmatrix} = \begin{bmatrix} 3.0 & 3.0 & 2.4 \end{bmatrix}$$

Sigmoid Operation on layer 2

$$\text{Sigmoid} = 1 / (1 + e^{-x})$$



$$\begin{bmatrix} h2_{out1} & h2_{out2} & h2_{out3} \end{bmatrix} = \begin{bmatrix} 1/(1 + e^{-h2_{in1}}) & 1/(1 + e^{-h2_{in2}}) & 1/(1 + e^{-h2_{in3}}) \end{bmatrix}$$

$$\begin{bmatrix} h2_{out1} & h2_{out2} & h2_{out3} \end{bmatrix} = \begin{bmatrix} 1/(1 + e^{-3.0}) & 1/(1 + e^{-3.0}) & 1/(1 + e^{-2.4}) \end{bmatrix}$$

$$\begin{bmatrix} h2_{out1} & h2_{out2} & h2_{out3} \end{bmatrix} = \begin{bmatrix} 0.952 & 0.952 & 0.916 \end{bmatrix}$$

Layer 3 Matrix Operation

$$= [h_{out1} \quad h_{out2} \quad h_{out3}] \times \begin{bmatrix} W_{k1l1} & W_{k1l2} & W_{k1l3} \\ W_{k2l1} & W_{k2l2} & W_{k2l3} \\ W_{k3l1} & W_{k3l2} & W_{k3l3} \end{bmatrix} + [b_{l1} \quad b_{l2} \quad b_{l3}]$$

$$= [0.952 \quad 0.952 \quad 0.916] \times \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.4 & 0.6 & 0.8 \\ 0.1 & 0.9 & 0.7 \end{bmatrix} + [1.0 \quad 1.0 \quad 1.0]$$

$$[0.7 \quad 1.9 \quad 1.7] + [1 \quad 1 \quad 1]$$

$$[o_{in1} \quad o_{in2} \quad o_{in3}] = [1.7 \quad 2.9 \quad 2.7]$$

$$\frac{e^{1.7}}{e^{1.7} + e^{2.9} + e^{2.7}} \quad \bigg| \quad \frac{e^{2.9}}{e^{1.7} + e^{2.9} + e^{2.7}} \quad \frac{e^{2.7}}{e^{1.7} + e^{2.9} + e^{2.7}}$$

Softmax operation on layer 3

$$\text{Softmax} = e^{o_{ina}} / (\sum_{a=1}^3 e^{o_{ina}})$$

$$\begin{bmatrix} o_{out1} & o_{out2} & o_{out3} \end{bmatrix} = \begin{bmatrix} e^{o_{ina}} / (\sum_{a=1}^3 e^{o_{ina}}) & e^{o_{ina}} / (\sum_{a=1}^3 e^{o_{ina}}) & e^{o_{ina}} / (\sum_{a=1}^3 e^{o_{ina}}) \end{bmatrix}$$

$$[o_{out1} \quad o_{out2} \quad o_{out3}] = [0.1420 \quad 0.4717 \quad 0.3862] \sim [0 \quad 1 \quad 0]$$

$$\begin{bmatrix} 0.142 \\ 0.471 \\ 0.386 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

The actual output is [0.0 1.0 0.0] but we got [0.1420 0.4717 0.3862]

We have done the forward pass

Compare the predicted result to the actual result. Measure the generated error.

Logistic Loss

$$\text{Log Loss} = -y_i * \log(o_{outi}) - (1 - y_i) * \log(1 - o_{outi})$$

$$= \sum_{i=1}^3 -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

$$\begin{aligned}
 E_1 &= -y_1 * \log(o_{out1}) - (1 - y_1) * \log(1 - o_{out1}) \\
 &= -0 * \log(0.1420) - (1 - 0) * \log(1 - 0.1420) \\
 &= -0 * \log(0.1420) - 1 * \log(1 - 0.1420) \\
 &= -\log(0.858) \\
 &= +0.0665 \\
 E_2 &= -y_2 * \log(o_{out2}) - (1 - y_2) * \log(1 - o_{out2}) \\
 &= -1 * \log(0.4717) - (1 - 1) * \log(1 - 0.4717) \\
 &= -1 * \log(0.4717) - 0 \\
 &= -\log(0.4717) \\
 &= +0.3263 \\
 E_3 &= -y_3 * \log(o_{out3}) - (1 - y_3) * \log(1 - o_{out3}) \\
 &= -0 * \log(0.3862) - (1 - 0) * \log(1 - 0.3862) \\
 &= -0 * \log(0.3862) - 1 * \log(1 - 0.3862) \\
 &= -\log(0.6138) \\
 &= +0.2119
 \end{aligned}$$

Σ

Cost Function

$$\begin{aligned}
 &= -\frac{1}{N} \left[\sum_{i=1}^N y_i * \log(o_{outi}) + (1 - y_i) * \log(1 - o_{outi}) \right] \\
 &= -\frac{1}{3} \left[\sum_{i=1}^3 y_i * \log(o_{outi}) + (1 - y_i) * \log(1 - o_{outi}) \right] \\
 &= \frac{0.0665 + 0.3263 + 0.2119}{3} \\
 &= 0.2015
 \end{aligned}$$

Feed Backward Neural Network: Back-Propagation: from left to right, the error is back-Propagated. Update the weights according to how much they are responsible for the error. The Learning rate decides how much we update the weights.

Back Propagating the error — (Hidden Layer 2 — Output Layer) Weights

Derivative of cross entropy

$$\frac{\partial E_i}{\partial o_{outi}} = \frac{-y_i * \log(o_{outi}) - (1 - y_i) * \log(1 - o_{outi})}{\partial o_{outi}}$$

$$\frac{\partial E_i}{\partial o_{outi}} = -y_i * (1/o_{outi}) - (1 - y_i) * (1/(1 - o_{outi}))$$

$$\begin{bmatrix} \frac{\partial E_1}{\partial o_{out1}} \\ \frac{\partial E_2}{\partial o_{out2}} \\ \frac{\partial E_3}{\partial o_{out3}} \end{bmatrix} = \begin{bmatrix} -y_1 * (1/o_{out1}) - (1 - y_1) * (1/(1 - o_{out1})) \\ -y_2 * (1/o_{out2}) - (1 - y_2) * (1/(1 - o_{out2})) \\ -y_3 * (1/o_{out3}) - (1 - y_3) * (1/(1 - o_{out3})) \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial E_1}{\partial o_{out1}} \\ \frac{\partial E_2}{\partial o_{out2}} \\ \frac{\partial E_3}{\partial o_{out3}} \end{bmatrix} = \begin{bmatrix} -1.1655 \\ -2.1199 \\ -1.6291 \end{bmatrix}$$

Derivative of Softmax layer

$$\frac{\partial o_{out1}}{\partial o_{in1}} = \frac{\partial (e^{o_{in1}} / (e^{o_{in1}} + e^{o_{in2}} + e^{o_{in3}}))}{\partial o_{in1}}$$

$$\begin{bmatrix} \frac{\partial o_{out1}}{\partial o_{in1}} \\ \frac{\partial o_{out2}}{\partial o_{in2}} \\ \frac{\partial o_{out3}}{\partial o_{in3}} \end{bmatrix} = \begin{bmatrix} (e^{o_{in1}} * (e^{o_{in2}} + e^{o_{in3}})) / (e^{o_{in1}} + e^{o_{in2}} + e^{o_{in3}})^2 \\ (e^{o_{in2}} * (e^{o_{in1}} + e^{o_{in3}})) / (e^{o_{in1}} + e^{o_{in2}} + e^{o_{in3}})^2 \\ (e^{o_{in3}} * (e^{o_{in1}} + e^{o_{in2}})) / (e^{o_{in1}} + e^{o_{in2}} + e^{o_{in3}})^2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial o_{out1}}{\partial o_{in1}} \\ \frac{\partial o_{out2}}{\partial o_{in2}} \\ \frac{\partial o_{out3}}{\partial o_{in3}} \end{bmatrix} = \begin{bmatrix} 0.1218 \\ 0.2492 \\ 0.2370 \end{bmatrix}$$

$$\frac{\partial o_{in1}}{\partial W_{k1l1}} = \frac{\partial((h2_{out1} * W_{k1l1}) + (h2_{out2} * W_{k2l1}) + (h2_{out3} * W_{k3l1}) + b_{l1})}{\partial W_{k1l1}}$$

$$\frac{\partial o_{in1}}{\partial W_{k1l1}} = h2_{out1}$$

$$\frac{\partial o_{in1}}{\partial W_{k2l1}} = \frac{\partial((h2_{out1} * W_{k1l1}) + (h2_{out2} * W_{k2l1}) + (h2_{out3} * W_{k3l1}) + b_{l1})}{\partial W_{k2l1}}$$

$$\frac{\partial o_{in1}}{\partial W_{k2l1}} = h2_{out2}$$

$$\frac{\partial o_{in1}}{\partial W_{k3l1}} = \frac{\partial((h2_{out1} * W_{k1l1}) + (h2_{out2} * W_{k2l1}) + (h2_{out3} * W_{k3l1}) + b_{l1})}{\partial W_{k3l1}}$$

$$\frac{\partial o_{in1}}{\partial W_{k3l1}} = h2_{out3}$$

$$\frac{\partial o_{in2}}{\partial W_{k1l2}} = \frac{\partial((h2_{out1} * W_{k1l2}) + (h2_{out2} * W_{k2l2}) + (h2_{out3} * W_{k3l2}) + b_{l2})}{\partial W_{k1l2}}$$

$$\frac{\partial o_{in2}}{\partial W_{k1l2}} = h2_{out1}$$

$$\frac{\partial o_{in2}}{\partial W_{k2l2}} = \frac{\partial((h2_{out1} * W_{k1l2}) + (h2_{out2} * W_{k2l2}) + (h2_{out3} * W_{k3l2}) + b_{l2})}{\partial W_{k2l2}}$$

$$\frac{\partial o_{in2}}{\partial W_{k2l2}} = h2_{out2}$$

$$\frac{\partial o_{in2}}{\partial W_{k3l2}} = \frac{\partial((h2_{out1} * W_{k1l2}) + (h2_{out2} * W_{k2l2}) + (h2_{out3} * W_{k3l2}) + b_{l2})}{\partial W_{k3l2}}$$

$$\frac{\partial o_{in2}}{\partial W_{k3l2}} = h2_{out3}$$

$$\frac{\partial o_{in3}}{\partial W_{k1l3}} = \frac{\partial((h2_{out1} * W_{k1l3}) + (h2_{out2} * W_{k2l3}) + (h2_{out3} * W_{k3l3}) + b_{l3})}{\partial W_{k1l3}}$$

$$\frac{\partial o_{in3}}{\partial W_{k1l3}} = h2_{out1}$$

$$\frac{\partial o_{in3}}{\partial W_{k2l3}} = \frac{\partial((h2_{out1} * W_{k1l3}) + (h2_{out2} * W_{k2l3}) + (h2_{out3} * W_{k3l3}) + b_{l3})}{\partial W_{k2l3}}$$

$$\frac{\partial o_{in3}}{\partial W_{k2l3}} = h2_{out2}$$

$$\frac{\partial o_{in3}}{\partial W_{k3l3}} = \frac{\partial((h2_{out1} * W_{k1l3}) + (h2_{out2} * W_{k2l3}) + (h2_{out3} * W_{k3l3}) + b_{l3})}{\partial W_{k3l3}}$$

$$\frac{\partial o_{in3}}{\partial W_{k3l3}} = h2_{out3}$$

$$\frac{\partial E_1}{\partial W_{k1l1}} = \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial W_{k1l1}}$$

$$\frac{\partial E_1}{\partial W_{k2l1}} = \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial W_{k2l1}}$$

$$\frac{\partial E_1}{\partial W_{k3l1}} = \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial W_{k3l1}}$$

$$\frac{\partial E_2}{\partial W_{k1l2}} = \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial W_{k1l2}}$$

$$\frac{\partial E_2}{\partial W_{k2l2}} = \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial W_{k2l2}}$$

$$\frac{\partial E_2}{\partial W_{k3l2}} = \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial W_{k3l2}}$$

$$\frac{\partial E_3}{\partial W_{k1l3}} = \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial W_{k1l3}}$$

$$\frac{\partial E_3}{\partial W_{k2l3}} = \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial W_{k2l3}}$$

$$\frac{\partial E_3}{\partial W_{k3l3}} = \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial W_{k3l3}}$$

$$\partial W_{kl} = \begin{bmatrix} \frac{\partial E_1}{\partial W_{k1l1}} & \frac{\partial E_2}{\partial W_{k1l2}} & \frac{\partial E_3}{\partial W_{k1l3}} \\ \frac{\partial E_1}{\partial W_{k2l1}} & \frac{\partial E_2}{\partial W_{k2l2}} & \frac{\partial E_3}{\partial W_{k2l3}} \\ \frac{\partial E_1}{\partial W_{k3l1}} & \frac{\partial E_2}{\partial W_{k3l2}} & \frac{\partial E_3}{\partial W_{k3l3}} \end{bmatrix}$$

$$\partial W_{kl} = \begin{bmatrix} \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial W_{k1l1}} & \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial W_{k1l2}} & \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial W_{k1l3}} \\ \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial W_{k2l1}} & \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial W_{k2l2}} & \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial W_{k2l3}} \\ \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial W_{k3l1}} & \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial W_{k3l2}} & \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial W_{k3l3}} \end{bmatrix}$$

$$\partial W_{kl} = \begin{bmatrix} -1.1655 * 0.1218 * 0.952 & -2.1199 * 0.2492 * 0.952 & -1.6291 * 0.2370 * 0.952 \\ -1.1655 * 0.1218 * 0.952 & -2.1199 * 0.2492 * 0.952 & -1.6291 * 0.2370 * 0.952 \\ -1.1655 * 0.1218 * 0.916 & -2.1199 * 0.2492 * 0.916 & -1.6291 * 0.2370 * 0.916 \end{bmatrix}$$

$$\partial W_{kl} = \begin{bmatrix} \partial W_{k1l1} & \partial W_{k1l2} & \partial W_{k1l3} \\ \partial W_{k2l1} & \partial W_{k2l2} & \partial W_{k2l3} \\ \partial W_{k3l1} & \partial W_{k3l2} & \partial W_{k3l3} \end{bmatrix}$$

$$\partial W_{kl} = \begin{bmatrix} -0.1351 & -0.5029 & -0.3675 \\ -0.1351 & -0.5029 & -0.3675 \\ -0.1300 & -0.4839 & -0.3536 \end{bmatrix}$$

$$W'_{kl} = \begin{bmatrix} W_{k1l1} - (lr * \partial W_{k1l1}) & W_{k1l2} - (lr * \partial W_{k1l2}) & W_{k1l3} - (lr * \partial W_{k1l3}) \\ W_{k2l1} - (lr * \partial W_{k2l1}) & W_{k2l2} - (lr * \partial W_{k2l2}) & W_{k2l3} - (lr * \partial W_{k2l3}) \\ W_{k3l1} - (lr * \partial W_{k3l1}) & W_{k3l2} - (lr * \partial W_{k3l2}) & W_{k3l3} - (lr * \partial W_{k3l3}) \end{bmatrix}$$

$$= \begin{bmatrix} 0.2 - (0.01 * -0.1351) & 0.5 - (0.01 * -0.5029) & 0.3 - (0.01 * -0.3675) \\ 0.4 - (0.01 * -0.1351) & 0.6 - (0.01 * -0.5029) & 0.8 - (0.01 * -0.3675) \\ 0.1 - (0.01 * -0.1300) & 0.9 - (0.01 * -0.4839) & 0.7 - (0.01 * -0.3536) \end{bmatrix}$$

$$= \begin{bmatrix} -0.2013 & -0.5050 & -0.3036 \\ -0.4013 & -0.6050 & -0.8036 \\ -0.1013 & -0.9048 & -0.7035 \end{bmatrix}$$

Back Propagating the error — (Hidden Layer 1 — Hidden Layer 2) Weights

Derivative of Sigmoid Layer

$$\frac{\partial h2_{out1}}{\partial h2_{in1}} = \frac{\partial Sigmoid(h2_{in1})}{\partial h2_{in1}}$$

$$\frac{\partial h2_{out1}}{\partial h2_{in1}} = Sigmoid(h2_{in1}) * (1 - Sigmoid(h2_{in1}))$$

$$\begin{bmatrix} \frac{\partial h2_{out1}}{\partial h2_{in1}} \\ \frac{\partial h2_{out2}}{\partial h2_{in2}} \\ \frac{\partial h2_{out3}}{\partial h2_{in3}} \end{bmatrix} = \begin{bmatrix} Sigmoid(h2_{in1}) * (1 - Sigmoid(h2_{in1})) \\ Sigmoid(h2_{in2}) * (1 - Sigmoid(h2_{in2})) \\ Sigmoid(h2_{in3}) * (1 - Sigmoid(h2_{in3})) \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial h2_{out1}}{\partial h2_{in1}} \\ \frac{\partial h2_{out2}}{\partial h2_{in2}} \\ \frac{\partial h2_{out3}}{\partial h2_{in3}} \end{bmatrix} = \begin{bmatrix} 0.0451 \\ 0.0451 \\ 0.0762 \end{bmatrix}$$

$$\frac{\partial h2_{in1}}{\partial W_{j1k1}} = \frac{\partial ((h1_{out1} * W_{j1k1}) + (h1_{out2} * W_{j2k1}) + (h1_{out3} * W_{j3k1}) + b_{k1})}{\partial W_{j1k1}}$$

$$\frac{\partial h_{2_{in1}}}{\partial W_{j1k1}} = h_{1_{out1}}$$

$$\frac{\partial h_{2_{in1}}}{\partial W_{j2k1}} = \frac{\partial((h_{1_{out1}} * W_{j1k1}) + (h_{1_{out2}} * W_{j2k1}) + (h_{1_{out3}} * W_{j3k1}) + b_{k1})}{\partial W_{j2k1}}$$

$$\frac{\partial h_{2_{in1}}}{\partial W_{j2k1}} = h_{1_{out2}}$$

$$\frac{\partial h_{2_{in1}}}{\partial W_{j3k1}} = \frac{\partial((h_{1_{out1}} * W_{j1k1}) + (h_{1_{out2}} * W_{j2k1}) + (h_{1_{out3}} * W_{j3k1}) + b_{k1})}{\partial W_{j3k1}}$$

$$\frac{\partial h_{2_{in1}}}{\partial W_{j3k1}} = h_{1_{out3}}$$

$$\frac{\partial h_{2_{in2}}}{\partial W_{j1k2}} = \frac{\partial((h_{1_{out1}} * W_{j1k2}) + (h_{1_{out2}} * W_{j2k2}) + (h_{1_{out3}} * W_{j3k2}) + b_{k2})}{\partial W_{j1k2}}$$

$$\frac{\partial h_{2_{in2}}}{\partial W_{j1k2}} = h_{1_{out1}}$$

$$\frac{\partial h_{2_{in2}}}{\partial W_{j2k2}} = \frac{\partial((h_{1_{out1}} * W_{j1k2}) + (h_{1_{out2}} * W_{j2k2}) + (h_{1_{out3}} * W_{j3k2}) + b_{k2})}{\partial W_{j2k2}}$$

$$\frac{\partial h_{2_{in2}}}{\partial W_{j2k2}} = h_{1_{out2}}$$

$$\frac{\partial h_{2_{in2}}}{\partial W_{j3k2}} = \frac{\partial((h_{1_{out1}} * W_{j1k2}) + (h_{1_{out2}} * W_{j2k2}) + (h_{1_{out3}} * W_{j3k2}) + b_{k2})}{\partial W_{j3k2}}$$

$$\frac{\partial h_{2_{in2}}}{\partial W_{j3k2}} = h_{1_{out3}}$$

$$\frac{\partial h_{2_{in3}}}{\partial W_{j1k3}} = \frac{\partial((h_{1_{out1}} * W_{j1k3}) + (h_{1_{out2}} * W_{j2k3}) + (h_{1_{out3}} * W_{j3k3}) + b_{k3})}{\partial W_{j1k3}}$$

$$\frac{\partial h_{2_{in3}}}{\partial W_{j1k3}} = h_{1_{out1}}$$

$$\frac{\partial h_{2_{in3}}}{\partial W_{j2k3}} = \frac{\partial((h_{1_{out1}} * W_{j1k3}) + (h_{1_{out2}} * W_{j2k3}) + (h_{1_{out3}} * W_{j3k3}) + b_{k3})}{\partial W_{j2k3}}$$

$$\frac{\partial h_{2_{in3}}}{\partial W_{j2k3}} = h_{1_{out2}}$$

$$\frac{\partial h_{2_{in3}}}{\partial W_{j3k3}} = \frac{\partial((h_{1_{out1}} * W_{j1k3}) + (h_{1_{out2}} * W_{j2k3}) + (h_{1_{out3}} * W_{j3k3}) + b_{k3})}{\partial W_{j3k3}}$$

$$\frac{\partial h_{2_{in3}}}{\partial W_{j3k3}} = h_{1_{out3}}$$

$$\frac{\partial E_{total}}{\partial W_{j1k1}} = \frac{\partial E_{total}}{\partial h_{2_{out1}}} * \frac{\partial h_{2_{out1}}}{\partial h_{2_{in1}}} * \frac{\partial h_{2_{in1}}}{\partial W_{j1k1}}$$

$$\frac{\partial E_{total}}{\partial W_{j2k1}} = \frac{\partial E_{total}}{\partial h_{2_{out1}}} * \frac{\partial h_{2_{out1}}}{\partial h_{2_{in1}}} * \frac{\partial h_{2_{in1}}}{\partial W_{j2k1}}$$

$$\frac{\partial E_{total}}{\partial W_{j3k1}} = \frac{\partial E_{total}}{\partial h_{2_{out1}}} * \frac{\partial h_{2_{out1}}}{\partial h_{2_{in1}}} * \frac{\partial h_{2_{in1}}}{\partial W_{j3k1}}$$

$$\frac{\partial E_{total}}{\partial W_{j1k2}} = \frac{\partial E_{total}}{\partial h_{2_{out2}}} * \frac{\partial h_{2_{out2}}}{\partial h_{2_{in2}}} * \frac{\partial h_{2_{in2}}}{\partial W_{j1k2}}$$

$$\frac{\partial E_{total}}{\partial W_{j2k2}} = \frac{\partial E_{total}}{\partial h_{2_{out2}}} * \frac{\partial h_{2_{out2}}}{\partial h_{2_{in2}}} * \frac{\partial h_{2_{in2}}}{\partial W_{j2k2}}$$

$$\frac{\partial E_{total}}{\partial W_{j3k2}} = \frac{\partial E_{total}}{\partial h2_{out2}} * \frac{\partial h2_{out2}}{\partial h2_{in2}} * \frac{\partial h2_{in2}}{\partial W_{j3k2}}$$

$$\frac{\partial E_{total}}{\partial W_{j1k3}} = \frac{\partial E_{total}}{\partial h2_{out3}} * \frac{\partial h2_{out3}}{\partial h2_{in3}} * \frac{\partial h2_{in3}}{\partial W_{j1k3}}$$

$$\frac{\partial E_{total}}{\partial W_{j2k3}} = \frac{\partial E_{total}}{\partial h2_{out3}} * \frac{\partial h2_{out3}}{\partial h2_{in3}} * \frac{\partial h2_{in3}}{\partial W_{j2k3}}$$

$$\frac{\partial E_{total}}{\partial W_{j3k3}} = \frac{\partial E_{total}}{\partial h2_{out3}} * \frac{\partial h2_{out3}}{\partial h2_{in3}} * \frac{\partial h2_{in3}}{\partial W_{j3k3}}$$

$$\partial W_{jk} = \begin{bmatrix} \frac{\partial E_{total}}{\partial W_{j1k1}} & \frac{\partial E_{total}}{\partial W_{j1k2}} & \frac{\partial E_{total}}{\partial W_{j1k3}} \\ \frac{\partial E_{total}}{\partial W_{j2k1}} & \frac{\partial E_{total}}{\partial W_{j2k2}} & \frac{\partial E_{total}}{\partial W_{j2k3}} \\ \frac{\partial E_{total}}{\partial W_{j3k1}} & \frac{\partial E_{total}}{\partial W_{j3k2}} & \frac{\partial E_{total}}{\partial W_{j3k3}} \end{bmatrix}$$

$$\partial W_{jk} = \begin{bmatrix} \frac{\partial E_{total}}{\partial h2_{out1}} * \frac{\partial h2_{out1}}{\partial h2_{in1}} * \frac{\partial h2_{in1}}{\partial W_{j1k1}} & \frac{\partial E_{total}}{\partial h2_{out2}} * \frac{\partial h2_{out2}}{\partial h2_{in2}} * \frac{\partial h2_{in2}}{\partial W_{j1k2}} & \frac{\partial E_{total}}{\partial h2_{out3}} * \frac{\partial h2_{out3}}{\partial h2_{in3}} * \frac{\partial h2_{in3}}{\partial W_{j1k3}} \\ \frac{\partial E_{total}}{\partial h2_{out1}} * \frac{\partial h2_{out1}}{\partial h2_{in1}} * \frac{\partial h2_{in1}}{\partial W_{j2k1}} & \frac{\partial E_{total}}{\partial h2_{out2}} * \frac{\partial h2_{out2}}{\partial h2_{in2}} * \frac{\partial h2_{in2}}{\partial W_{j2k2}} & \frac{\partial E_{total}}{\partial h2_{out3}} * \frac{\partial h2_{out3}}{\partial h2_{in3}} * \frac{\partial h2_{in3}}{\partial W_{j2k3}} \\ \frac{\partial E_{total}}{\partial h2_{out1}} * \frac{\partial h2_{out1}}{\partial h2_{in1}} * \frac{\partial h2_{in1}}{\partial W_{j3k1}} & \frac{\partial E_{total}}{\partial h2_{out2}} * \frac{\partial h2_{out2}}{\partial h2_{in2}} * \frac{\partial h2_{in2}}{\partial W_{j3k2}} & \frac{\partial E_{total}}{\partial h2_{out3}} * \frac{\partial h2_{out3}}{\partial h2_{in3}} * \frac{\partial h2_{in3}}{\partial W_{j3k3}} \end{bmatrix}$$

$$\frac{\partial E_{total}}{\partial h2_{out1}} = \frac{\partial E_1}{\partial h2_{out1}} * \frac{\partial E_2}{\partial h2_{out1}} * \frac{\partial E_3}{\partial h2_{out1}}$$

$$\frac{\partial E_1}{\partial h2_{out1}} = \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial h2_{out1}}$$

$$\frac{\partial E_2}{\partial h2_{out1}} = \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial h2_{out1}}$$

$$\frac{\partial E_3}{\partial h2_{out1}} = \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial h2_{out1}}$$

$$\frac{\partial E_{total}}{\partial h2_{out2}} = \frac{\partial E_1}{\partial h2_{out2}} * \frac{\partial E_2}{\partial h2_{out2}} * \frac{\partial E_3}{\partial h2_{out2}}$$

$$\frac{\partial E_1}{\partial h2_{out2}} = \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial h2_{out2}}$$

$$\frac{\partial E_2}{\partial h2_{out2}} = \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial h2_{out2}}$$

$$\frac{\partial E_3}{\partial h2_{out2}} = \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial h2_{out2}}$$

$$\frac{\partial E_{total}}{\partial h2_{out3}} = \frac{\partial E_1}{\partial h2_{out3}} * \frac{\partial E_2}{\partial h2_{out3}} * \frac{\partial E_3}{\partial h2_{out3}}$$

$$\frac{\partial E_1}{\partial h2_{out3}} = \frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial o_{in1}}{\partial h2_{out3}}$$

$$\frac{\partial E_2}{\partial h2_{out3}} = \frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial o_{in2}}{\partial h2_{out3}}$$

$$\frac{\partial E_3}{\partial h2_{out3}} = \frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial o_{in3}}{\partial h2_{out3}}$$

$$\begin{bmatrix} \frac{\partial E_{total}}{\partial h2_{out1}} \\ \frac{\partial E_{total}}{\partial h2_{out2}} \\ \frac{\partial E_{total}}{\partial h2_{out3}} \end{bmatrix} = \begin{bmatrix} \left(\frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial h2_{out1}}{\partial h2_{out1}} \right) + \left(\frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial h2_{out1}}{\partial h2_{out1}} \right) + \left(\frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial h2_{out1}}{\partial h2_{out1}} \right) \\ \left(\frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial h2_{out2}}{\partial h2_{out2}} \right) + \left(\frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial h2_{out2}}{\partial h2_{out2}} \right) + \left(\frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial h2_{out2}}{\partial h2_{out2}} \right) \\ \left(\frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * \frac{\partial h2_{out3}}{\partial h2_{out3}} \right) + \left(\frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * \frac{\partial h2_{out3}}{\partial h2_{out3}} \right) + \left(\frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * \frac{\partial h2_{out3}}{\partial h2_{out3}} \right) \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial o_{in1}}{\partial h2_{out1}} & \frac{\partial o_{in2}}{\partial h2_{out1}} & \frac{\partial o_{in3}}{\partial h2_{out1}} \\ \frac{\partial o_{in1}}{\partial h2_{out2}} & \frac{\partial o_{in2}}{\partial h2_{out2}} & \frac{\partial o_{in3}}{\partial h2_{out2}} \\ \frac{\partial o_{in1}}{\partial h2_{out3}} & \frac{\partial o_{in2}}{\partial h2_{out3}} & \frac{\partial o_{in3}}{\partial h2_{out3}} \end{bmatrix} = \begin{bmatrix} W_{k1l1} & W_{k1l2} & W_{k1l3} \\ W_{k2l1} & W_{k2l2} & W_{k2l3} \\ W_{k3l1} & W_{k3l2} & W_{k3l3} \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial E_{total}}{\partial h2_{out1}} \\ \frac{\partial E_{total}}{\partial h2_{out2}} \\ \frac{\partial E_{total}}{\partial h2_{out3}} \end{bmatrix} = \begin{bmatrix} \left(\frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * W_{k1l1} \right) + \left(\frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * W_{k1l2} \right) + \left(\frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * W_{k1l3} \right) \\ \left(\frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * W_{k2l1} \right) + \left(\frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * W_{k2l2} \right) + \left(\frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * W_{k2l3} \right) \\ \left(\frac{\partial E_1}{\partial o_{out1}} * \frac{\partial o_{out1}}{\partial o_{in1}} * W_{k3l1} \right) + \left(\frac{\partial E_2}{\partial o_{out2}} * \frac{\partial o_{out2}}{\partial o_{in2}} * W_{k3l2} \right) + \left(\frac{\partial E_3}{\partial o_{out3}} * \frac{\partial o_{out3}}{\partial o_{in3}} * W_{k3l3} \right) \end{bmatrix}$$

$$= \begin{bmatrix} (-1.1655 * 0.1218 * 0.2) + (-2.1199 * 0.2492 * 0.5) + (-1.6291 * 0.2370 * 0.3) \\ (-1.1655 * 0.1218 * 0.4) + (-2.1199 * 0.2492 * 0.6) + (-1.6291 * 0.2370 * 0.8) \\ (-1.1655 * 0.1218 * 0.1) + (-2.1199 * 0.2492 * 0.9) + (-1.6291 * 0.2370 * 0.7) \end{bmatrix}$$

$$= \begin{bmatrix} (-0.0283) + (-0.2641) + (-0.1158) \\ (-0.0567) + (-0.3169) + (-0.3088) \\ (-0.0141) + (-0.4754) + (-0.2702) \end{bmatrix}$$

$$= \begin{bmatrix} -0.4082 \\ -0.6824 \\ -0.7597 \end{bmatrix}$$

$$\partial W_{jk} = \begin{bmatrix} -0.4082 * 0.0451 * 1.2 & -0.6824 * 0.0451 * 1.2 & -0.7597 * 0.0762 * 1.2 \\ -0.4082 * 0.0451 * 1.6 & -0.6824 * 0.0451 * 1.6 & -0.7597 * 0.0762 * 1.6 \\ -0.4082 * 0.0451 * 1.5 & -0.6824 * 0.0451 * 1.5 & -0.7597 * 0.0762 * 1.5 \end{bmatrix}$$

$$\partial W_{jk} = \begin{bmatrix} -0.0220 & -0.0369 & -0.0694 \\ -0.0294 & -0.0492 & -0.0926 \\ -0.0276 & -0.0461 & -0.0868 \end{bmatrix}$$

$$W'_{jk} = \begin{bmatrix} W_{j1k1} - (lr * \partial W_{j1k1}) & W_{j1k2} - (lr * \partial W_{j1k2}) & W_{j1k3} - (lr * \partial W_{j1k3}) \\ W_{j2k1} - (lr * \partial W_{j2k1}) & W_{j2k2} - (lr * \partial W_{j2k2}) & W_{j2k3} - (lr * \partial W_{j2k3}) \\ W_{j3k1} - (lr * \partial W_{j3k1}) & W_{j3k2} - (lr * \partial W_{j3k2}) & W_{j3k3} - (lr * \partial W_{j3k3}) \end{bmatrix}$$

$$W'_{jk} = \begin{bmatrix} 0.4 - (0.01 * -0.0220) & 0.6 - (0.01 * -0.0369) & 0.1 - (0.01 * -0.0694) \\ 0.2 - (0.01 * -0.0294) & 0.7 - (0.01 * -0.0492) & 0.3 - (0.01 * -0.0926) \\ 0.8 - (0.01 * -0.0276) & 0.1 - (0.01 * -0.0461) & 0.5 - (0.01 * -0.0868) \end{bmatrix}$$

$$W'_{jk} = \begin{bmatrix} -0.4002 & -0.6003 & -0.1006 \\ -0.2002 & -0.7004 & -0.3009 \\ -0.8002 & -0.1004 & -0.5008 \end{bmatrix}$$

Back Propagating the error — (Input Layer — Hidden Layer 1)

Derivation of Relu Layer

$$\frac{\partial h1_{out1}}{\partial h1_{in1}} = \frac{\partial Relu(h1_{in1})}{\partial h1_{in1}}$$

$$\begin{bmatrix} \frac{\partial h1_{out1}}{\partial h1_{in1}} \\ \frac{\partial h1_{out2}}{\partial h1_{in2}} \\ \frac{\partial h1_{out3}}{\partial h1_{in3}} \end{bmatrix} = \begin{bmatrix} 1.00 \\ 1.00 \\ 1.00 \end{bmatrix}$$

$$\frac{\partial h1_{in1}}{\partial W_{i1j1}} = \frac{\partial ((I_{out1} * W_{i1j1}) + (I_{out2} * W_{i2j1}) + (I_{out3} * W_{i3j1}) + b_{j1})}{\partial W_{i1j1}}$$

$$\frac{\partial I_{in1}}{\partial W_{i1j1}} = I_{out1}$$

$$\frac{\partial h1_{in1}}{\partial W_{i2j1}} = \frac{\partial ((I_{out1} * W_{i1j1}) + (I_{out2} * W_{i2j1}) + (I_{out3} * W_{i3j1}) + b_{j1})}{\partial W_{i2j1}}$$

$$\frac{\partial I_{in1}}{\partial W_{i2j1}} = I_{out2}$$

$$\frac{\partial h1_{in1}}{\partial W_{i3j1}} = \frac{\partial((I_{out1} * W_{i1j1}) + (I_{out2} * W_{i2j1}) + (I_{out3} * W_{i3j1}) + b_{j1})}{\partial W_{i3j1}}$$

$$\frac{\partial I_{in1}}{\partial W_{i3j1}} = I_{out3}$$

$$\frac{\partial h1_{in1}}{\partial W_{i1j2}} = \frac{\partial((I_{out1} * W_{i1j2}) + (I_{out2} * W_{i2j2}) + (I_{out3} * W_{i3j2}) + b_{j2})}{\partial W_{i1j2}}$$

$$\frac{\partial I_{in1}}{\partial W_{i1j2}} = I_{out1}$$

$$\frac{\partial h1_{in1}}{\partial W_{i2j2}} = \frac{\partial((I_{out1} * W_{i1j2}) + (I_{out2} * W_{i2j2}) + (I_{out3} * W_{i3j2}) + b_{j2})}{\partial W_{i2j2}}$$

$$\frac{\partial I_{in1}}{\partial W_{i2j2}} = I_{out2}$$

$$\frac{\partial h1_{in1}}{\partial W_{i3j2}} = \frac{\partial((I_{out1} * W_{i1j2}) + (I_{out2} * W_{i2j2}) + (I_{out3} * W_{i3j2}) + b_{j2})}{\partial W_{i3j2}}$$

$$\frac{\partial I_{in1}}{\partial W_{i3j2}} = I_{out3}$$

$$\frac{\partial h1_{in1}}{\partial W_{i1j3}} = \frac{\partial((I_{out1} * W_{i1j3}) + (I_{out2} * W_{i2j3}) + (I_{out3} * W_{i3j3}) + b_{j3})}{\partial W_{i1j3}}$$

$$\frac{\partial I_{in1}}{\partial W_{i1j3}} = I_{out1}$$

$$\frac{\partial h1_{in1}}{\partial W_{i2j3}} = \frac{\partial((I_{out1} * W_{i1j3}) + (I_{out2} * W_{i2j3}) + (I_{out3} * W_{i3j3}) + b_{j3})}{\partial W_{i2j3}}$$

$$\frac{\partial I_{in1}}{\partial W_{i2j3}} = I_{out2}$$

$$\frac{\partial h1_{in1}}{\partial W_{i3j3}} = \frac{\partial((I_{out1} * W_{i1j3}) + (I_{out2} * W_{i2j3}) + (I_{out3} * W_{i3j3}) + b_{j3})}{\partial W_{i3j3}}$$

$$\frac{\partial I_{in1}}{\partial W_{i3j3}} = I_{out3}$$

$$\partial W_{ij} = \begin{bmatrix} \frac{\partial E_{total}}{\partial W_{i1j1}} & \frac{\partial E_{total}}{\partial W_{i1j2}} & \frac{\partial E_{total}}{\partial W_{i1j3}} \\ \frac{\partial E_{total}}{\partial W_{i2j1}} & \frac{\partial E_{total}}{\partial W_{i2j2}} & \frac{\partial E_{total}}{\partial W_{i2j3}} \\ \frac{\partial E_{total}}{\partial W_{i3j1}} & \frac{\partial E_{total}}{\partial W_{i3j2}} & \frac{\partial E_{total}}{\partial W_{i3j3}} \end{bmatrix}$$

$$\partial W_{ij} = \begin{bmatrix} \frac{\partial E_{total}}{\partial h1_{out1}} * \frac{\partial h1_{out1}}{\partial h1_{in1}} * \frac{\partial h1_{in1}}{\partial W_{i1j1}} & \frac{\partial E_{total}}{\partial h1_{out2}} * \frac{\partial h1_{out2}}{\partial h1_{in2}} * \frac{\partial h1_{in2}}{\partial W_{i1j2}} & \frac{\partial E_{total}}{\partial h1_{out3}} * \frac{\partial h1_{out3}}{\partial h1_{in3}} * \frac{\partial h1_{in3}}{\partial W_{i1j3}} \\ \frac{\partial E_{total}}{\partial h1_{out1}} * \frac{\partial h1_{out1}}{\partial h1_{in1}} * \frac{\partial h1_{in1}}{\partial W_{i2j1}} & \frac{\partial E_{total}}{\partial h1_{out2}} * \frac{\partial h1_{out2}}{\partial h1_{in2}} * \frac{\partial h1_{in2}}{\partial W_{i2j2}} & \frac{\partial E_{total}}{\partial h1_{out3}} * \frac{\partial h1_{out3}}{\partial h1_{in3}} * \frac{\partial h1_{in3}}{\partial W_{i2j3}} \\ \frac{\partial E_{total}}{\partial h1_{out1}} * \frac{\partial h1_{out1}}{\partial h1_{in1}} * \frac{\partial h1_{in1}}{\partial W_{i3j1}} & \frac{\partial E_{total}}{\partial h1_{out2}} * \frac{\partial h1_{out2}}{\partial h1_{in2}} * \frac{\partial h1_{in2}}{\partial W_{i3j2}} & \frac{\partial E_{total}}{\partial h1_{out3}} * \frac{\partial h1_{out3}}{\partial h1_{in3}} * \frac{\partial h1_{in3}}{\partial W_{i3j3}} \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial E_{total}}{\partial h1_{out1}} \\ \frac{\partial E_{total}}{\partial h1_{out2}} \\ \frac{\partial E_{total}}{\partial h1_{out3}} \end{bmatrix} = \begin{bmatrix} \frac{\partial E_{total}}{\partial h2_{out1}} * \frac{\partial h2_{out1}}{\partial h2_{in1}} * \frac{\partial h2_{in1}}{\partial h1_{out1}} \\ \frac{\partial E_{total}}{\partial h2_{out2}} * \frac{\partial h2_{out2}}{\partial h2_{in2}} * \frac{\partial h2_{in2}}{\partial h1_{out2}} \\ \frac{\partial E_{total}}{\partial h2_{out3}} * \frac{\partial h2_{out3}}{\partial h2_{in3}} * \frac{\partial h2_{in3}}{\partial h1_{out3}} \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial E_{total}}{\partial h1_{out1}} \\ \frac{\partial E_{total}}{\partial h1_{out2}} \\ \frac{\partial E_{total}}{\partial h1_{out3}} \end{bmatrix} = \begin{bmatrix} \frac{\partial E_{total}}{\partial h2_{out1}} * \frac{\partial h2_{out1}}{\partial h2_{in1}} * W_{j1k1} \\ \frac{\partial E_{total}}{\partial h2_{out2}} * \frac{\partial h2_{out2}}{\partial h2_{in2}} * W_{j2k2} \\ \frac{\partial E_{total}}{\partial h2_{out3}} * \frac{\partial h2_{out3}}{\partial h2_{in3}} * W_{j3k3} \end{bmatrix}$$

$$= \begin{bmatrix} -0.4082 * 0.0451 * 0.4 \\ -0.6824 * 0.0451 * 0.7 \\ -0.7597 * 0.0762 * 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} -0.007363 \\ -0.021543 \\ -0.028944 \end{bmatrix}$$

$$W_{ij} = \begin{bmatrix} \frac{\partial E_{total}}{\partial h1_{out1}} * \frac{\partial h1_{out1}}{\partial h1_{in1}} * \frac{\partial h1_{in1}}{\partial W_{i1j1}} & \frac{\partial E_{total}}{\partial h1_{out2}} * \frac{\partial h1_{out2}}{\partial h1_{in2}} * \frac{\partial h1_{in2}}{\partial W_{i1j2}} & \frac{\partial E_{total}}{\partial h1_{out3}} * \frac{\partial h1_{out3}}{\partial h1_{in3}} * \frac{\partial h1_{in3}}{\partial W_{i1j3}} \\ \frac{\partial E_{total}}{\partial h1_{out1}} * \frac{\partial h1_{out1}}{\partial h1_{in1}} * \frac{\partial h1_{in1}}{\partial W_{i2j1}} & \frac{\partial E_{total}}{\partial h1_{out2}} * \frac{\partial h1_{out2}}{\partial h1_{in2}} * \frac{\partial h1_{in2}}{\partial W_{i2j2}} & \frac{\partial E_{total}}{\partial h1_{out3}} * \frac{\partial h1_{out3}}{\partial h1_{in3}} * \frac{\partial h1_{in3}}{\partial W_{i2j3}} \\ \frac{\partial E_{total}}{\partial h1_{out1}} * \frac{\partial h1_{out1}}{\partial h1_{in1}} * \frac{\partial h1_{in1}}{\partial W_{i3j1}} & \frac{\partial E_{total}}{\partial h1_{out2}} * \frac{\partial h1_{out2}}{\partial h1_{in2}} * \frac{\partial h1_{in2}}{\partial W_{i3j2}} & \frac{\partial E_{total}}{\partial h1_{out3}} * \frac{\partial h1_{out3}}{\partial h1_{in3}} * \frac{\partial h1_{in3}}{\partial W_{i3j3}} \end{bmatrix}$$

$$W_{ij} = \begin{bmatrix} -0.007363 * 1 * 0.1 & -0.021543 * 1 * 0.1 & -0.028944 * 1 * 0.1 \\ -0.007363 * 1 * 0.3 & -0.021543 * 1 * 0.3 & -0.028944 * 1 * 0.3 \\ -0.007363 * 1 * 0.5 & -0.021543 * 1 * 0.5 & -0.028944 * 1 * 0.5 \end{bmatrix}$$

$$W_{ij} = \begin{bmatrix} -0.0007363 & -0.0021543 & -0.0028944 \\ -0.0022089 & -0.0064629 & -0.0086832 \\ -0.0036815 & -0.0107715 & -0.014472 \end{bmatrix}$$

$$W'_{ij} = \begin{bmatrix} W_{i1j1} - (lr * \partial W_{i1j1}) & W_{i1j2} - (lr * \partial W_{i1j2}) & W_{i1j3} - (lr * \partial W_{i1j3}) \\ W_{i2j1} - (lr * \partial W_{i2j1}) & W_{i2j2} - (lr * \partial W_{i2j2}) & W_{i2j3} - (lr * \partial W_{i2j3}) \\ W_{i3j1} - (lr * \partial W_{i3j1}) & W_{i3j2} - (lr * \partial W_{i3j2}) & W_{i3j3} - (lr * \partial W_{i3j3}) \end{bmatrix}$$

$$W'_{ij} =$$

$$\begin{bmatrix} 0.2 - (0.01 * -0.0007363) & 0.4 - (0.01 * -0.0021543) & 0.5 - (0.01 * -0.0028944) \\ 0.3 - (0.01 * -0.0022089) & 0.7 - (0.01 * -0.0064629) & 0.1 - (0.01 * -0.0086832) \\ 0.2 - (0.01 * -0.0036815) & 0.6 - (0.01 * -0.0107715) & 0.9 - (0.01 * -0.014472) \end{bmatrix}$$

$$W'_{ij} = \begin{bmatrix} 0.20007 & 0.4003 & 0.5007 \\ 0.30016 & 0.7001 & 0.1007 \\ 0.20078 & 0.6008 & 0.9009 \end{bmatrix}$$